

EMERSON EIMS BUILDING SUITE PRO

PDF UK

Location: London, UK
Total Area: 180 m² | Units: 1 | Storeys: 3
Architectural Style: Modern
Date: 27 April 2026

PROJECT INFORMATION	
Project ID	8031e6ae-951f-40fe-a3f6-3a4309d759f1
Location	London, UK
Total Area	180 m²
Units / Apartments	1
Storeys	3
Bedrooms per Unit	4
Foundation Type	Raft
Style	Modern
Report Generated	2026-04-27 03:03

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VALIDATION ENGINE — AUDIT TRAIL

■ 1 BLOCKER(S) — DO NOT SUBMIT

• UK statutory IDs missing: Plot / cadastral reference; Architect (must be registered with ARB / RIBA); Structural Engineer (must be registered with ICE / IStructE); Quantity Surveyor (must be registered with RICS). Required for permit submission to Local Authority Building Control / Approved Inspector.

1 auto-correction(s) applied with code citation:

Field	Original	Auto-corrected to	Reason / Code citation
Foundation	(unspecified)	raft	3-storey × 180 m² → raft (BS 8004 §7 / EN 1997-1)

Compliance Status: **BLOCKED — DO NOT SUBMIT**

1. DESIGN BRIEF & DESIGN CRITERIA

Item	Value
Project name	PDF UK
Location	London, UK
Total floor area	180.0 m²
No. of storeys / floor count	3 (single source of truth)
Dwelling units	1
Bedrooms / unit	4
Architectural style	Modern
Foundation	RAFT
Currency	GBP
Plot / LR No.	— REQUIRED —
Coordinates (lat,lng)	-1.2921,36.8219
Title Deed	—
Architect (BORAQS)	— REQUIRED —
Structural Engineer (EBK)	— REQUIRED —
Quantity Surveyor (BORAQS)	— REQUIRED —
Client	—

Design loads (BS EN 1991-1-1 Cat A — residential):

Action	Value (kN/m²)	Reference
Permanent action g_k (slab + finishes + partitions)	7.00	EC1 Table 6.1
Variable action q_k	2.00	EC1 Table 6.2 Cat A
Design ULS $w = 1.35 g_k + 1.5 q_k$	12.45	EC0 §6.4.3.2
Design SLS $w = g_k + q_k$	9.00	EC0 §6.5.3

2. STRUCTURAL DESIGN CALCULATIONS

Calculations to BS EN 1992-1-1 (EC2) using $f_{ck} = 30$ MPa, $f_{yk} = 500$ MPa, cover per exposure class XC1 internal / XC3 external. All members hand-verified against simplified Eurocode 2 design rules; final detailing to be reviewed by the Structural Engineer of Record (EBK-registered) before issue for construction.

2.1 Slab design (one/two-way RC slab)

Parameter	Value
Short span L_x	7.07 m
Long span L_y	8.49 m
Behaviour	Two-way
Slab thickness h	270 mm (span/depth = 26)
Effective depth d	239.0 mm
Cover	25 mm (XC1)
Design moment M	38.6 kN·m / m
As required	391.0 mm ² / m
As minimum (EC2 §9.3.1.1)	351.0 mm ² / m
As provided	391.0 mm ² / m
Reinforcement	T12 @ 275 c/c BW

2.2 Beam design (simply-supported RC beam)

Parameter	Value
Clear span L	4.24 m
Beam section $b \times h$	225 × 350 mm
Effective depth d	304.0 mm
Cover	30 mm
Service load w	45.99 kN/m
Design moment M	103.5 kN·m
Design shear V	97.6 kN
As required (tension)	824.0 mm ²
Main steel	5 no. T16 bot · 3 no. T16 top
Shear links	T8 @ 225 c/c

2.3 Column design (axially loaded RC column)

Parameter	Value
Axial load N (ULS)	855.0 kN
Section	225 × 225 mm
Concrete grade	C25/30 ($f_{ck} = 30$ MPa)
Steel grade	$f_{yk} = 500$ MPa
Steel ratio ρ	0.020 (2.0 %)
As required	1012.0 mm ²
Main steel	6 no. T16
Links	T8 @ 192 c/c

2.4 Foundation design (per BS 8004 / EN 1997-1)

Parameter	Value
Foundation type	RAFT
Plan size	Whole footprint
Thickness	350 mm
Concrete grade	C25/30
Cover	75 mm (XC2 ground contact)
Safe bearing pressure (presumed)	150 kPa
Column load to foundation	855.0 kN
Reinforcement	T16 @ 200 c/c BW top + bottom

■ The bearing pressure shown is a BS 8004 Table 1 *presumed* value pending a site investigation. A signed Geotechnical Investigation Report (boreholes, SPT, allowable bearing capacity) must be obtained from a geotechnical engineer before construction. EIMS records this gap honestly rather than fabricating SPT values.

3. BAR BENDING SCHEDULE (BS 8666)

Schedule of all reinforcement bars with mark, diameter, shape code, length per bar, number off, total length and weight. Steel grade 500B. Bend radii and bar shapes comply with BS 8666:2020 Table 2. Quantities to be verified by the Engineer before cutting and bending.

Bar Mark	Dia.	Shape Code	Length per bar (m)	No. off	Total length (m)	Weight (kg)
S1	T12	00	13.72	80	1097.31	974.4
B1-bot	T16	00	4.74	64	303.53	479.3
B1-top	T16	37	4.74	32	151.76	239.6
B1-link	T8	51	0.85	288	244.80	96.7
C1	T16	00	3.50	48	168.00	265.3
C1-link	T8	51	0.60	144	86.40	34.1
F1	T12	00	3.00	96	288.00	255.7
					TOTAL	2345.1

Total reinforcement steel: 2,345 kg $\approx$ 2.35 tonnes

4. MEP DESIGN — ELECTRICAL, PLUMBING & DRAINAGE

Electrical design per IEC 60364 / KS IEC 60364. Plumbing & sanitary drainage per BS EN 12056-2 and Kenya NCC Building Code SEHS. Septic tank to NEMA EIA Regulations 2003.

4.1 Electrical load schedule

Load description	Value
Lighting (LED) @ 5 W/m ²	900 W
Small power sockets @ 25 W/m ²	4500 W
Cooker / hob (per unit)	6000 W
Water heaters	6000 W
Air conditioning (split units)	6000 W
TOTAL CONNECTED LOAD	23,400 W
Diversity factor	0.65
DESIGN MAXIMUM DEMAND	15,210 W (16.0 kVA)
Main breaker (single phase 230 V)	70 A

4.2 Final-circuit schedule

Circuit	Load	Cable	Protection	RCD
L1-Ln (Lighting)	900 W	1.5 mm ² T+E	10 A B-curve MCB	—
P1-Pn (Sockets)	4500 W	2.5 mm ² T+E	20 A B-curve MCB	30 mA RCD
C (Cooker)	6000 W	6 mm ² T+E	32 A B-curve MCB	—
W (2× water heater)	6000 W	4 mm ² T+E	25 A B-curve MCB	30 mA RCD
A1-A4 (AC splits)	6000 W	2.5 mm ² T+E	16 A C-curve MCB	—

4.3 Single-line diagram (text representation)

KPLC supply ■ kWh meter ■ Main isolator 70 A ■ Main DB (Type-B SPD)
■ Lighting circuits L1...Ln (1.5 mm² · 10 A)
■ Socket circuits P1...Pn (2.5 mm² · 20 A · 30 mA RCD)
■ Cooker C (6 mm² · 32 A)
■ Water heaters W (4 mm² · 25 A · 30 mA RCD)
■ AC splits A1...An (2.5 mm² · 16 A)
■ Earth busbar ■ copper earth-rod ≤ 1 Ω

4.4 Plumbing fixture schedule

Fixture	Qty
Water Closet (WC)	2
Wash-Hand Basin	2
Shower / Bath	2
Kitchen Sink	1
Laundry Sink	1
External Bib Tap	2

4.5 Drainage gradient table (BS EN 12056-2)

Pipe	Slope range	Min fall
100 mm soil pipe (WC)	1:40 to 1:80	≥ 1.00 m fall per 1 m
75 mm waste	1:40	0.025 m / m

50 mm waste	1:40	0.025 m / m
32 mm WHB / sink	1:20	0.050 m / m

4.6 Septic tank & soak-pit sizing

Parameter	Value
Design occupancy	8 persons (BS 8233 — bedrooms × 2)
Daily wastewater flow (180 L/p/d)	1,440 L/day
Septic tank volume (3-day retention)	4.32 m ³
Suggested tank dimensions	2.5 × 1.2 × 1.44 m (L×W×D)
Soak-pit area (porous soil, 30 L/m ² /d)	48.0 m ²
Soak-pit diameter (circular)	7.82 m

■ Final septic-system geometry must be confirmed by an on-site percolation test. A NEMA Environmental Impact Assessment (EIA) project report is required for systems > 50 persons or sites near surface watercourses.

5. BILL OF QUANTITIES (KQS-2025)

Contractor-grade BOQ in GBP, structured by trade per the Kenya Quantity Surveyors Standard Method of Measurement. Rates are KQS-2025 indicative urban rates; tenderers shall price all items. All quantities to be verified on site.

A — PRELIMINARIES

Description	Unit	Qty	Rate	Amount
Site mobilisation, hoarding, signage	item	1.00	£1,091	£1,091
Insurance & NCA Levy	item	1.00	£1,333	£1,333
Site office + storage	item	1.00	£576	£576
			Trade subtotal	£3,000

B — SUBSTRUCTURE

Description	Unit	Qty	Rate	Amount
Site clearance & strip topsoil	m²	234.00	£1	£170
Bulk excavation foundation trenches	m³	54.00	£4	£213
Hardcore filling 200 mm	m²	180.00	£5	£927
Lean concrete blinding 50 mm	m³	9.00	£85	£764
RC foundation C25/30	m³	54.00	£158	£8,509
			Trade subtotal	£10,583

C — SUPERSTRUCTURE

Description	Unit	Qty	Rate	Amount
RC columns C25/30	m³	4.25	£197	£838
RC beams C25/30	m³	16.04	£185	£2,964
RC suspended slabs C25/30	m³	145.80	£173	£25,184
Reinforcement bars (BBS)	kg	2,345.10	£1	£2,345
Formwork to soffits & sides	m²	996.54	£9	£8,757
200 mm machine-cut block walling	m²	482.99	£11	£5,269
Plaster both sides	m²	965.98	£7	£6,440
			Trade subtotal	£51,797

D — ROOFING

Description	Unit	Qty	Rate	Amount
Timber roof trusses & purlins	m²	180.00	£13	£2,400
0.6 mm box-profile coloured iron sheet	m²	189.00	£9	£1,661
9 mm gypsum ceiling on softwood brandering	m²	540.00	£11	£6,055
			Trade subtotal	£10,115

E — FINISHES

Description	Unit	Qty	Rate	Amount
Floor ceramic / porcelain tiling	m²	459.00	£15	£6,676
Wall tiling (wet areas)	m²	97.20	£13	£1,296
Internal paint (Crown / Plascon emulsion)	m²	772.79	£3	£2,435
External paint (weather-shield)	m²	386.39	£4	£1,686

			Trade subtotal	£12,094
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F — JOINERY

Description	Unit	Qty	Rate	Amount
Internal flush doors with hardware	each	7.00	£109	£764
External main doors hardwood	each	2.00	£255	£509
Aluminium-framed sliding windows	m²	27.00	£58	£1,555
			Trade subtotal	£2,827

G — ELECTRICAL

Description	Unit	Qty	Rate	Amount
Mains supply, DB, MCB, RCDs (per IEC 60364)	item	1.00	£1,000	£1,000
Lighting points & circuits (~900 W)	point	32.40	£13	£432
Socket outlets (~4500 W)	point	21.60	£15	£314
			Trade subtotal	£1,746

H — PLUMBING

Description	Unit	Qty	Rate	Amount
Cold + hot water reticulation, all fixtures	item	1.00	£1,727	£1,727
Septic tank + soak-pit (4.32 m³ + Ø7.82 m)	item	1.00	£1,182	£1,182
Stormwater drainage & rain-water disposal	item	1.00	£879	£879
			Trade subtotal	£3,788

Cost summary

Item	Amount
A — PRELIMINARIES	£3,000
B — SUBSTRUCTURE	£10,583
C — SUPERSTRUCTURE	£51,797
D — ROOFING	£10,115
E — FINISHES	£12,094
F — JOINERY	£2,827
G — ELECTRICAL	£1,746
H — PLUMBING	£3,788
Subtotal of works	£95,951
Contingency 7.5 %	£7,196
Professional fees 8 % (Architect + Engineer + QS)	£7,676
Sub-total before VAT	£110,823
VAT 16 %	£17,732
GRAND TOTAL	£128,555
Cost per m² (GBP)	£714 / m²

6. STATUTORY COMPLIANCE PACK

All approvals required under Kenyan law before construction. The contractor and professional team shall ensure each item is closed prior to mobilisation.

Approval	Authority / Statute	Documents required
1. County Planning Approval	County Govt (Building Plans Section)	Stamped architectural & structural drawings, location plan, ownership docs, NCA
2. NCA Project Registration	National Construction Authority Act 2011	NCA-1 form, contractor registration certificate, levy 0.5 % of contract sum.
3. NEMA Project Report (PR)	EMCA 1999, EIA Regs 2003	Required for projects > 5 ha or near watercourses; PR submitted to NEMA county
4. Public Health Approval	Public Health Act Cap. 242	Septic system & water supply review by County Public Health Office.
5. EPRA / KPLC Energy Connection	EPRA Regulations	Electrical drawings & wayleave application; final inspection by EPRA-registered C
6. Fire Safety Certificate	County Fire Marshal	Means-of-escape, extinguisher schedule, hydrant layout (NFPA 241 / Kenya Fire
7. DOSHS Notification	OSHA 2007 §41	Notification before commencement; site safety plan, induction registers, ISO 4500

6.1 Professional certification (signature pages)

Role	Name & Reg.	Signature & Stamp	Date
Architect (BORAQS)	— REQUIRED —	_____	Date: _____
Structural Engineer (EBK)	— REQUIRED —	_____	Date: _____
Quantity Surveyor (BORAQS)	— REQUIRED —	_____	Date: _____
Site Manager (NCA)	— to be appointed —	_____	Date: _____
Client	—	_____	Date: _____

I certify that the design has been carried out in accordance with the relevant Kenyan codes (NCA Act 2011, EMCA 1999, OSHA 2007) and applicable Eurocodes / British Standards, and that I take professional responsibility for the section of work above corresponding to my registration.

8. PROJECT SUMMARY

Phase phase_1: Site Analysis

water_table_depth	3.5
water_table_source	regional_estimate
water_table_note	Regional estimate from climate/soil model -- verify with on-site geotechnical su
soil_type	clay
vegetation_coverage	0.65
slope_angle	8.25
suitability_pct	90
nearby_structures	survey_required
land_use	residential
climate_zone	tropical
analysis_method	regional_data_model
data_source	EIMS Regional Database + GPS Coordinates

Phase phase_2: Geotechnical Design

bearing_capacity_kPa	150
soil_classification	CH
recommended_foundation	STRIP
settlement_mm	40
slope_stability	stable
water_table_depth	3.5
excavation_depth	2.8

Phase phase_3: Foundation Design

foundation_type	PAD
foundation_depth_m	1.65
footing_width_m	1.96
total_building_load_kN	1350.0
bearing_capacity_kPa	150
safety_factor	3.0
concrete_volume_m3	44.55
reinforcement_steel_kg	4009.5
soil_type	clay
water_table_factor	standard
design_standard	BS 8004 / Eurocode 7
bim_ready	True

Phase phase_4: Floor Plans

floor_count	3
total_area_sqm	180.0
bathrooms	3
rooms	{'bedrooms': 4, 'bathrooms': 3, 'living_areas': 1, 'kitchen': 1, 'extra_rooms':

style	modern
area_per_unit	180.0
building_code_compliance	True
area_efficiency_percent	80.0
cost_per_sqm_estimate	calculated_in_costing_phase

Phase phase_5: Electrical Design

total_connected_load_kW	9.38
max_demand_kW	5.63
diversity_factor	0.6
supply_type	single-phase 230V
main_current_A	24.5
lighting_points	12
socket_outlets	12
total_circuits	7
lighting_circuits	2
socket_circuits	2
fixed_appliance_circuits	3
distribution_boards	1

Phase phase_6: Plumbing Design

total_occupants	6
daily_demand_liters	900
peak_hourly_liters	90
total_fixture_units	34.5
main_supply_pipe_mm	32
cold_water_pipes	11
hot_water_pipes	7
waste_outlets	10
soil_stacks	1
hot_water_storage_liters	240
water_heater_kW	9.6
storage_tank_liters	1000

Phase phase_7: BOQ

material_items	77
total_material_cost	51882.33
currency	USD
items_by_category	{'mep_hvac': 5, 'electrical': 11, 'finishes': 18, 'plumbing': 14, 'external': 12}
items	[{'item_no': 1, 'description': 'Portland Cement (50kg)', 'unit': 'bag', 'quantit
calculation_basis	{'total_area_m2': 180.0, 'stories': 3, 'units': 1, 'bedrooms_per_unit': 4, 'foot
waste_factor_applied	True
method	quantity_takeoff_from_dimensions

Phase phase_8: Infrastructure

solar_system	{'panels_kw': 3.6, 'panels_count': 9, 'battery_kWh': 5.4, 'estimated_cost_usd':
borehole_system	{'depth_meters': 8, 'estimated_yield_lpm': 15, 'pump_kw': 3, 'estimated_cost_usd
grid_option	{'connection_cost_usd': 240, 'monthly_tariff_estimate_usd': 27}
recommendation	solar_grid_hybrid
estimated_annual_saving_usd	670.68

Phase phase_9: Landscape Design

lawn_area_m2	63.0
hardscape_area_m2	18.0
garden_area_m2	27.0
plants_count	216
irrigation_system	False
recommended_style	tropical_lush
estimated_cost_usd	621
maintenance_monthly_usd	12

Phase phase_10: Permits & Compliance

jurisdiction	UK
building_code	Building Regulations 2010
building_code_compliance	REQUIRES_VERIFICATION
permits_required	5
estimated_timeline_days	42
estimated_fees_usd	346
setback_compliance	{'front_m': 3, 'side_m': 1, 'rear_m': 3}
required_documents	['Architectural drawings (stamped)', 'Structural calculations (engineer-certifie

Phase phase_11: 3D Visualization

status	ready
model_endpoint	/api/drawings/3d-model

Phase phase_12: Live Costing

material_cost	40987.04
labor_cost	17521.96
equipment_cost	3278.96
overhead	7021.08
contingency	6880.9
subtotal	75689.95
contractor_profit	15137.99
total_project_cost	90827.94
cost_per_sqm	504.6
labor_multiplier	0.95
region	UK
currency	GBP

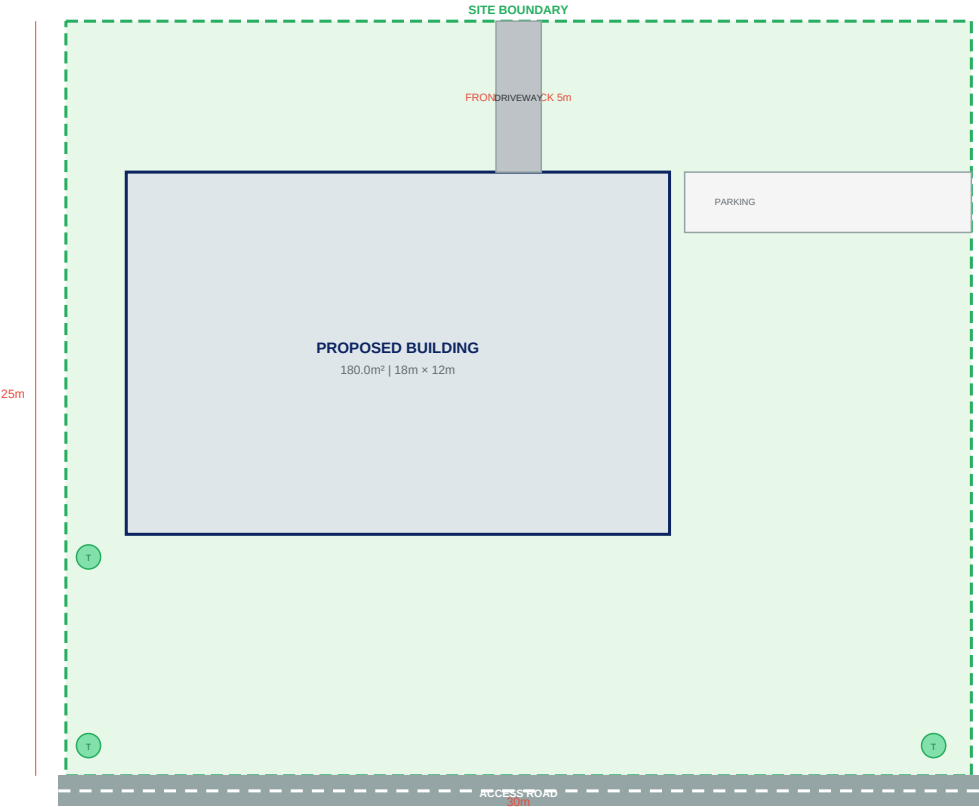
Phase phase_13: Integration Layer

status	complete
export_formats	['IFC2x3', 'FBX', 'DXF', 'PDF', 'XLSX']

9. SITE PLAN

Site layout showing building footprint, access roads, setbacks and landscaping. Area: 180 m²

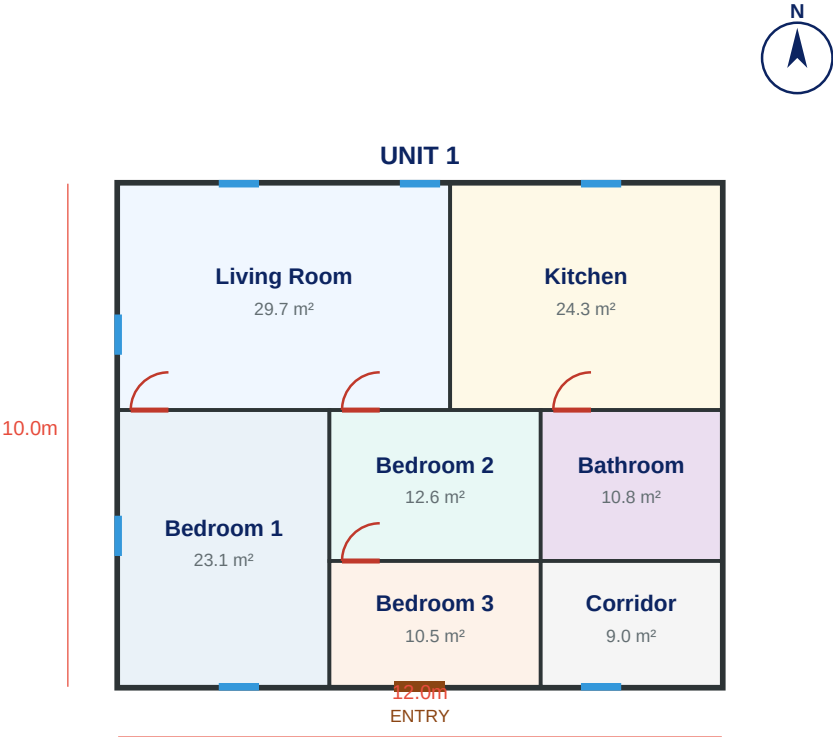
SITE PLAN - London, UK | Plot: 30m×25m



10. ARCHITECTURAL FLOOR PLAN

Ground floor layout — 1 unit(s), 4 bedrooms, showing room dimensions, doors, windows and circulation.

FLOOR PLAN - 180.0m² | 1 Units | 3 Storey(s)



EMERSON EIMS BUILDING SUITE PRO

Area: 180.0m² | 1 Units × 4 Bed

Floor Plan | Scale 1:50 | 2026-04-27 | Drawing: FP-001

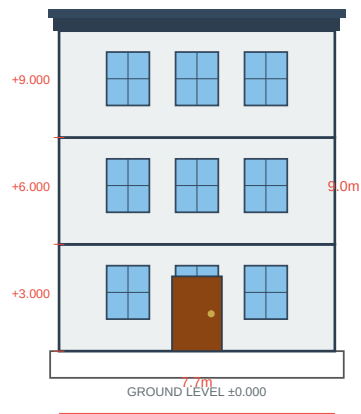
IFC Ready | BIM Level 2

11. BUILDING ELEVATIONS

4a. NORTH ELEVATION

Front view showing 3 storey(s), window/door openings, roof profile and floor levels.

NORTH ELEVATION - 3 Stories | Style: Modern

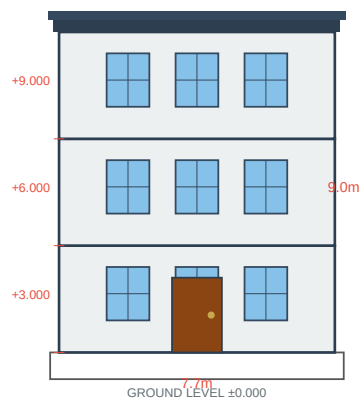


EMERSON EIMS | NORTH ELEVATION | Stories: 3 | Height: 9.0m
Scale 1:50 | 2026-04-27 | Drawing: EL-N01 | Style: Modern | IFC Ready

4b. EAST ELEVATION

Side view showing building depth, roof slope and storey heights.

EAST ELEVATION - 3 Stories | Style: Modern

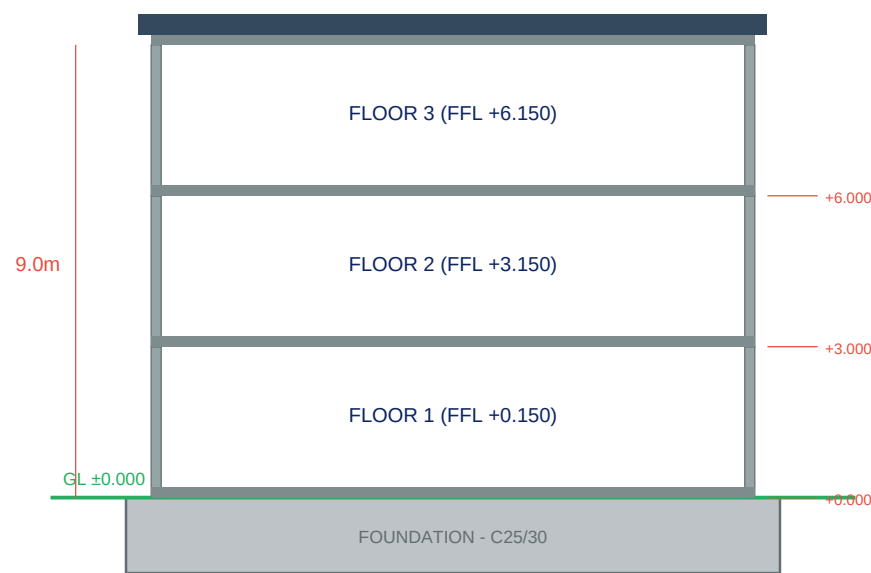


EMERSON EIMS | EAST ELEVATION | Stories: 3 | Height: 9.0m
Scale 1:50 | 2026-04-27 | Drawing: EL-E01 | Style: Modern | IFC Ready

12. BUILDING CROSS SECTION

Vertical cut through building showing floor slabs, storey heights (3 floors), foundation depth, roof structure and structural members.

CROSS SECTION A-A | 3 Stories



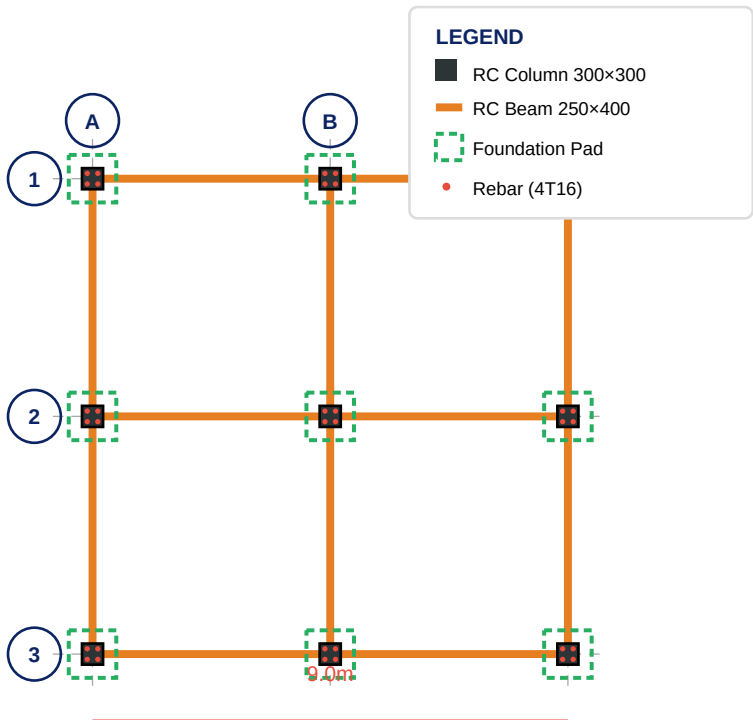
EMERSON EIMS | BUILDING SECTION A-A

2026-04-27 | SEC-001 | Scale 1:50

13. STRUCTURAL LAYOUT

Column grid, beam layout and foundation pads. Foundation type: Raft. All dimensions in metres. Concrete grade C25/30, Steel Y16.

STRUCTURAL LAYOUT - Raft Foundation | 3F | Grid 4.5m×4.5m



EMERSON EIMS | STRUCTURAL LAYOUT

Grid: 4.5m × 4.5m | Columns: 9 | 2026-04-27 | STR-001

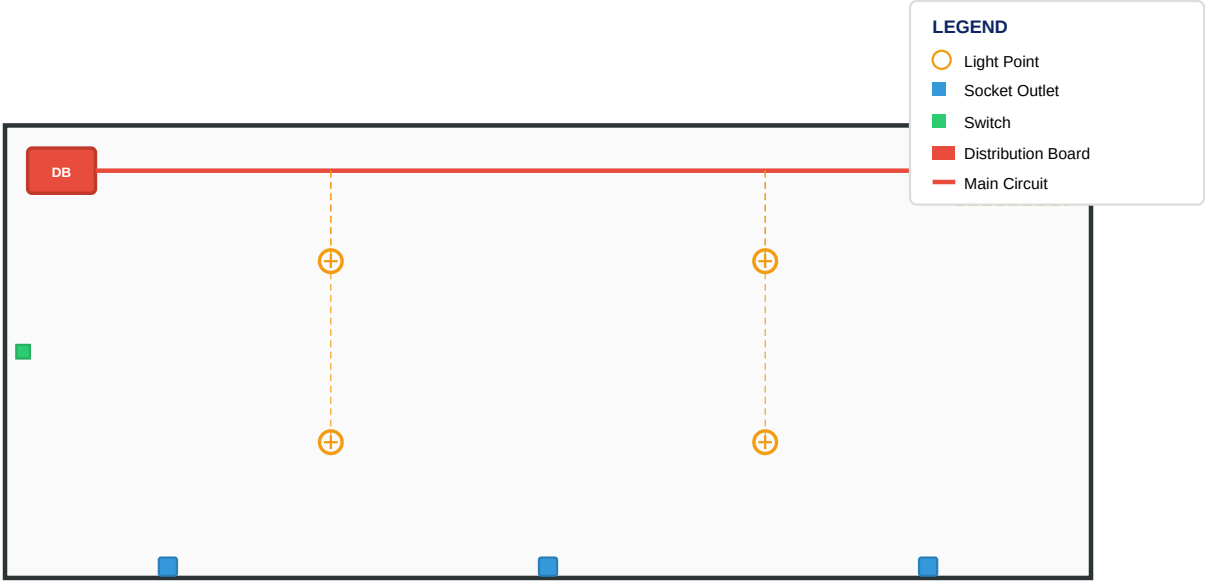
Concrete: C25/30 | Steel: Y16

Foundation: Raft | FOS ≥ 3.0

14. MEP LAYOUT — ELECTRICAL

Electrical distribution layout showing DB positions, circuit runs, socket/light points and earthing.

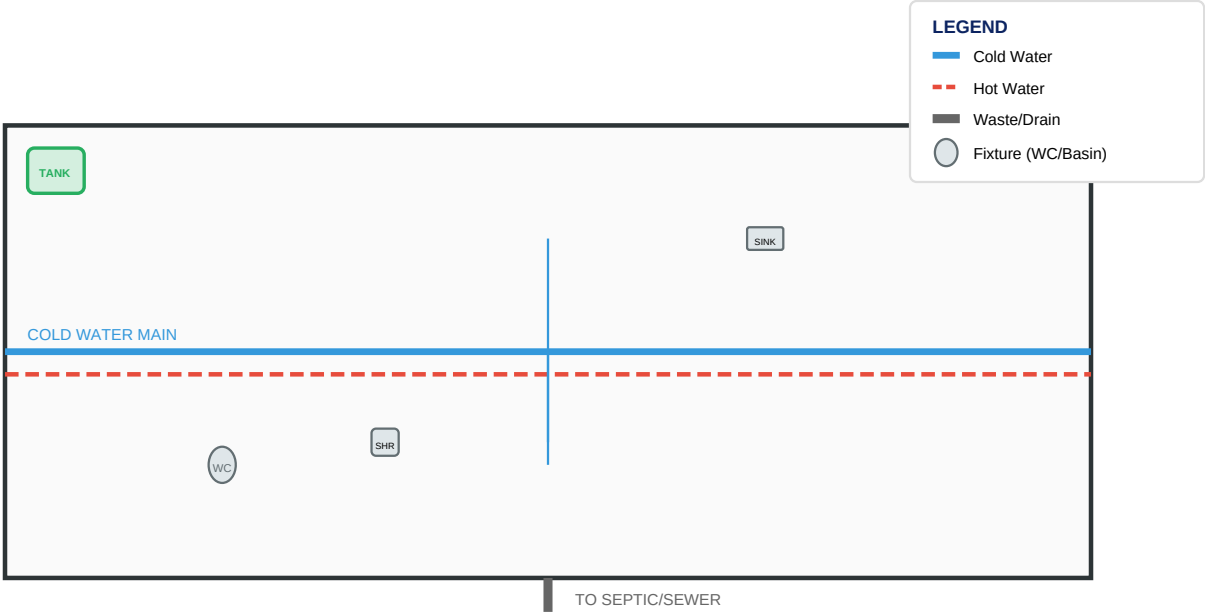
ELECTRICAL LAYOUT - 180.0m² | 1 Units



15. MEP LAYOUT — PLUMBING

Plumbing layout showing supply/drain runs, fixture locations, water heater and main shutoff valve.

PLUMBING LAYOUT - 180.0m² | 1 Units



16. COST SUMMARY

Item	Amount (GBP)
Material Cost	£40,987.04
Labor Cost	£17,521.96
Equipment Cost	£3,278.96
Overhead	£7,021.08
Contingency	£6,880.90
Subtotal	£75,689.95
Contractor Profit	£15,137.99
Total Project Cost	£90,827.94
Cost Per Sqm	£504.60
Labor Multiplier	£0.95
Fx From Usd	£0.79

This report was generated by EMERSON EIMS Building Suite Pro. All drawings are indicative and must be verified by a licensed engineer before construction.